

Using Optical Head-Mounted Displays for instant, contextualized feedback

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Abstract – Optical Head-Mounted Displays (OHMDs), such as Google Glass, have substantial potential in learning situations where contextualized, instantaneous feedback is beneficial. We present a study of a Google Glass-based application that allows instructors to give feedback to students while they are doing an in-class presentation. We examined students' perception of the usability of the system, the usefulness of the feedback, as well as its impact on students' self-efficacy. We found that students responded very positively to the feedback, reported no major usability issues, and overall agreed that their self-efficacy increased. The positive responses to this study support further research into the learning affordances of OHMDs to deliver contextualized, instant feedback.

Keywords – OHMDs, augmented reality, instant feedback, formative feedback, oral presentation skills

I. INTRODUCTION

The utilization of Optical Head-Mounted Displays (OHMDs), such as the Google Glass used in this study, has been explored in a wide variety of areas such as manufacturing, healthcare, and navigation; however, there is limited research on the use of OHMDs in education [1, 2]. OHMDs may offer unique benefits for students and teachers in giving instant feedback when compared to traditional methods [3, 4].

An example is the learning of oral presentation skills. The way presentation skills are commonly taught is by focusing on preparation, that is, students are instructed to plan their presentations carefully ahead of time and practice them frequently. However, there is wide recognition that presentations should be considered a social activity, mastery of which involves the ability to adapt to the audience and respond contextually to arising contingencies [5]. A presentation is for an audience, and, as such becoming proficient in presentations cannot be fully accomplished through improved preparation but is best served by contextualized instant feedback.

So far, very little research has examined the potential of OHMDs to deliver such feedback in a learning context. This study was conducted to obtain initial data on usability and self-reported learning. We employed semi-structured interviews and sentiment analysis, through which we rated the tone of the participants' answers [6]. The feedback centered on the oral presentation itself, targeting verbal and non-verbal behavior, including bodily behaviors which are critical for effective presentations.

II. RESEARCH QUESTIONS

1. How do the students perceive the Google Glass in terms of usability?
2. Do students find the instant feedback useful?
3. Do students perceive the feedback system as an effective means to improve their presentation skills?

III. METHOD

Data collection and analysis was based on the main constructs of usability studies: effectiveness, efficiency, and satisfaction [14]. In this context, we define effectiveness as the ease of use of the smart glass feedback system; and efficiency as the impact of the feedback on participants' behavior. In addition, we also collected data on students' self-efficacy, a critical factor in learning which affects persistency, and the likeliness to work harder [15]. The questions are reported in table 1.

A. The OHMD

We used a Google Glass Enterprise Edition 2, with a 640-pixel x 360-pixel RGB display just above the user's right eye. This device sits on the user's head using a wire-frame glasses mount. It runs Android Open-Source Project 8.1 (Oreo) A gallery app is included in the standard Glass apps. The app allowed to display pictures on the Glass prism. The display was controlled by a laptop connected to the Glass via Bluetooth.

B. Participants and procedure

We recruited and interviewed 4 graduate and 7 undergraduate students. Demographics: 9 identified as White, one as Latino and one as African American; three were females and 8 males. This sample size is generally considered sufficient to detect major usability issues, including those related to ease of use. During in-class presentations, students were given the Google Glass and asked to wear it during their presentation. They lasted about 8-10 minutes.

Feedback was given by selecting (tapping) on a card displayed on the laptop screen. The feedback card would then display on the Glass. Example of feedback cards were: *Speed up*, *Slow down*, *Eye Contact*, *Louder*, *Quieter*, *Good*, *Wrap-up*. Constructive feedback was printed on orange cards, supportive on green cards, and warning on red cards.

Table 1. Results of interview questions analyses with examples. Counts to be compared to $N = 11$.

Category	Questions	Answer types	Example Answers	Count
Usability of Feedback System and Google Glass	Tell me how you felt when you were wearing Google Glasses?	Easy-to-Use	“ I felt comfortable, it didn't feel like it was that noticeable. I wasn't too distracted by it.”	8
		Uncomfortable	“ I felt a little bit uncomfortable because I never used it before, but it was helpful, I liked it.”	3
	Are you distracted due to the instant feedback during the presentation?	Distracted	“ I think so. I tried to adapt to what it was saying to me, and follow those feedbacks, like ‘ speak up’ ‘ speak faster’ and ‘ eye contact’ .”	3
		Not Distracted	“ No, it didn't distract me at all. I thought it was comfortable to wear; it wasn't heavy or anything, it felt like my glasses.”	8
	When did you really feel comfortable during the presentation?	Comfortable	“ After I get used the glasses, I felt really comfortable at all times.”	11
Learning	Tell me a time when you did NOT feel comfortable during the presentation?	Comfortable	“ I can say at the beginning of the session, I tried to get used to it. It took me a minute to adopt reading the screen.”	10
		Somewhat Comfortable	“ Definitely at the start when I was getting used to it, But then again when I really saw that flash of red. It made me think of a stop sign, and it caught me off guard. It made me stumble a couple of times. That was mainly what tripped me up.”	1
	Did you change your behavior?	Yes	“ Definitely, yes! The feedback reminded me what should I do when speaking. The notifications helped me to stay on track.”	11
	Do you think working with Google Glasses could help grow your presentation skills? How so?	Yes	“ Yes, I think so. I think I would have to get used to it in order to use it to its full extent. I think it would help me with staying where I need to be in the presentation and doing those little things correctly to keep the audience engaged.”	11
	Do you recall the feedback shown?	Recall at least 3 out of 6 feedbacks	“ Yes, I know that there was ‘ go slower’ and ‘ talk louder’ . Those were kind of red(bad). There was also good, like ‘ more eye contact’ . Those were the ones I remember.”	11
	Please tell me how you felt receiving feedback from the instructor?	Engaging	“ It was actually great. I was waiting for the first feedback, and I really liked getting it. It made me feel safer, that is, I kind of had the sense that someone would tell me if I did something wrong.”	11

IV. RESULTS

We used a sentiment analysis approach to interpret the interview responses [6]. Student answers were classified into positive, neutral, and negative, adapted to the specific question (e.g., “I was distracted” is classified as a negative answer). All student answers were first classified independently by two researchers; classifications were then compared. There were no disagreements in the classifications. The answers so classified were counted to obtain cardinalities for each type of answer. Table 1 reports the interview questions, classification of student answers, examples for each, and count.

V. DISCUSSION

As can be seen in the high numbers of positive responses, students concur that the system is easy to use, that it is useful, and that they experience a greater sense of self-efficacy. It appears that knowing that they will be given feedback by their instructor and that they will be able to change their behavior in-vivo, they acquire a sense that they will learn while doing the presentation. The Glass’ capabilities to project private feedback is of relevance here. Public feedback is disruptive, and students may fear that their shortcoming are exposed with negative effects on their learning.

Overall, the study provided sufficient confirmation on the usability of the Glass as an instant feedback system. Student responses also gave us directions towards further improving the system, including not using the color orange (because it can be disruptive) and giving more positive feedback when student changed their behavior (e.g., make eye contact after having been given feedback to that effect).

The study also opened further lines of inquiry. Importantly, students talked spontaneously about the impact of the feedback (including anticipated feedback) on their communication apprehension. Communication apprehension is considered one major factor influencing performances and learning of giving presentations. They concurred that that apprehension was lower than it would have been without expecting or receiving feedback. The study wasn’t planned to examine this aspect but considered these responses a line of inquiry into this effect is highly recommended. Training of presentation skills commonly includes gauging students’ communication apprehension to tailor summative feedback and adapt training to students with high anxiety (using the widely used and well validated PRCA-24 survey). In future research we will submit that questionnaire to participants before and after a presentation to examine the impact of anticipating and receiving instant feedback.

Further, the aspect of immersion was not examined in this study. There is broad recognition that innovative technology, including AR technology, can increase students’ sense of immersion. Thus, we may observe a synergistic effect of the greater sense of immersion engendered by instant feedback.

VI. CONCLUSIONS

Although the OHMD technology has been available for almost a decade, there is limited research on utilizing such technology for learning. This study shows the potential of transforming how students are trained in presentation skills. Google Glasses represent the next stage of technological advancement in the classroom, but more research needs to be done to ensure that smart glasses reach their full educational potential. It is worth noting that presentations skills are critical for other users as well, especially teachers. Apprehension and lack of confidence toward one’s presentations skills are directly related to job satisfaction, and a factor in developing healthy student-teacher relationships [9]. Thus, the benefits of introducing OHMD-delivered instant feedback into presentation training are likely to go beyond the context in which this study was conducted.

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